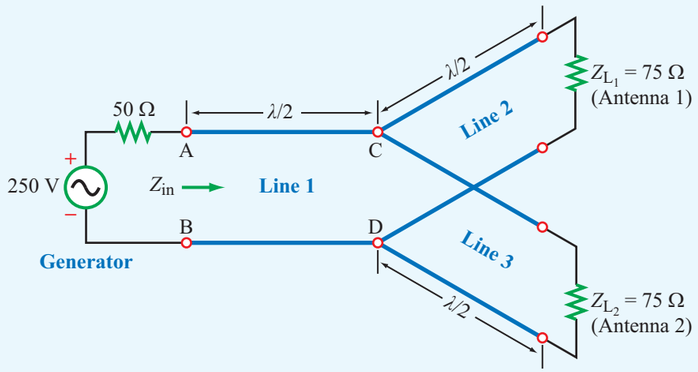


Figure P2.43 Antenna configuration for Problem 2.43.

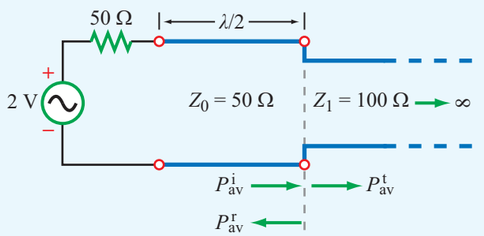


Figure P2.44 Circuit for Problem 2.44.

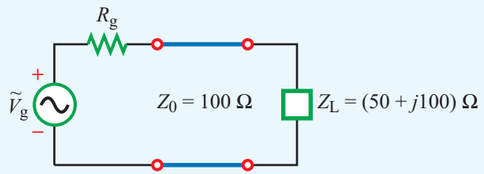


Figure P2.45 Circuit for Problem 2.45.

$$Z_L = (75 + j25)\ \Omega$$

is connected to a transmitter through a 50 Ω lossless transmission line. If under matched conditions (50 Ω load) the transmitter can deliver 20 W to the load, how much power can it deliver to the antenna? Assume that $Z_g = Z_0$.

Section 2-10: Smith Chart

2.47 Use the Smith chart to find the reflection coefficient corresponding to a load impedance of

(a) $Z_L = 3Z_0$

*(b)** $Z_L = (2 - j2)Z_0$

(c) $Z_L = -j2Z_0$

(d) $Z_L = 0$ (short circuit)

2.48 Repeat Problem 2.47 using Module 2.6.

2.49 Use the Smith chart to find the normalized load impedance corresponding to a reflection coefficient of

(a) $\Gamma = 0.5$

(b) $\Gamma = 0.5\angle 60°$

(c) $\Gamma = -1$

(d) $\Gamma = 0.3\angle -30°$

(e) $\Gamma = 0$

(f) $\Gamma = j$

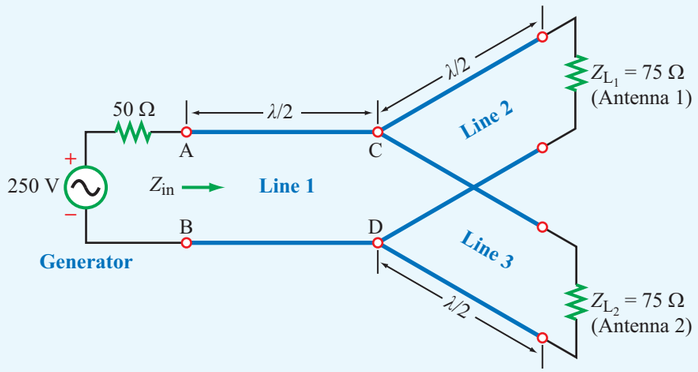


图 P2.43 习题2.43的天线连接。

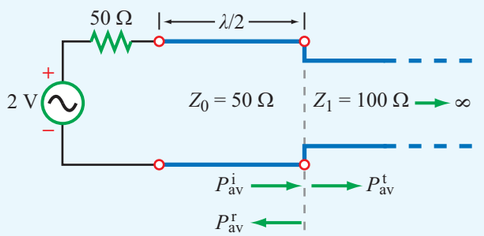


图 P2.44 习题2.44的电路。

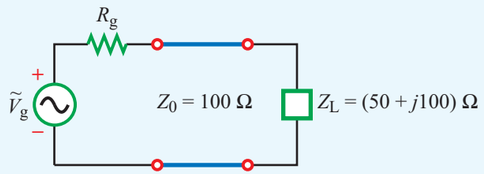


图 P2.45 习题2.45的电路。

$$Z_L = (75 + j25)\ \Omega$$

天线通过50 Ω无损传输线连接发射机。若匹配负载（50 Ω）时发射机可向负载提供20 W，接入该天线时可提供多少功率？设 Zg=Z0。

第2-10节 史密斯圆图

2.47 用史密斯圆图求下列负载阻抗对应的反射系数。

(a) $Z_L = 3Z_0$

*(b)** $Z_L = (2 - j2)Z_0$

(c) $Z_L = -j2Z_0$

(d) $Z_L = 0$（短路）

2.48 用模块2.6重做习题2.47。

2.49 用史密斯圆图求下列反射系数对应的归一化负载阻抗。

(a) $\Gamma = 0.5$

(b) $\Gamma = 0.5\angle 60°$

(c) $\Gamma = -1$

(d) $\Gamma = 0.3\angle -30°$

(e) $\Gamma = 0$

(f) $\Gamma = j$

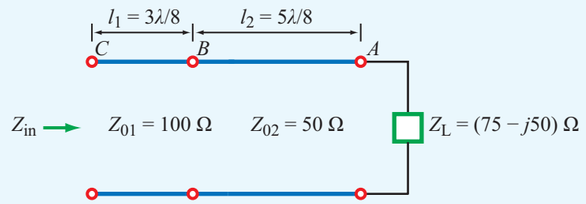


Figure P2.50 Circuit for Problem 2.50.

*2.50 Use the Smith chart to determine the input impedance Z_{in} of the two-line configuration shown in Fig. P2.50.

2.51 Repeat Problem 2.50 using Module 2.6.

*2.52 On a lossless transmission line terminated in a load $Z_L = 100\ \Omega$, the standing-wave ratio was measured to be 2.5. Use the Smith chart to find the two possible values of Z_0.

2.53 A lossless 50 Ω transmission line is terminated in a load with $Z_L = (50 + j25)\ \Omega$. Use the Smith chart to find the following:

(a) The reflection coefficient Γ.

*(b) The standing-wave ratio.

(c) The input impedance at 0.35λ from the load.

(d) The input admittance at 0.35λ from the load.

(e) The shortest line length for which the input impedance is purely resistive.

(f) The position of the first voltage maximum from the load.

2.54 Repeat Problem 2.53 using Module 2.6.

*2.55 A lossless 50 Ω transmission line is terminated in a short circuit. Use the Smith chart to determine:

(a) The input impedance at a distance 2.3λ from the load.

(b) The distance from the load at which the input admittance is $Y_{in} = -j0.04$ S.

2.56 Repeat Problem 2.55 using Module 2.6.

*2.57 Use the Smith chart to find y_L if $z_L = 1.5 - j0.7$.

2.58 A lossless 100 Ω transmission line $3\lambda/8$ in length is terminated in an unknown impedance. If the input impedance is $Z_{in} = -j2.5\ \Omega$,

(a) Use the Smith chart to find Z_L.

(b) Verify your results using Module 2.6.

2.59 A 75 Ω lossless line is 0.6λ long. If $S = 1.8$ and $\theta_r = -60°$, use the Smith chart to find $|\Gamma|$, Z_L, and Z_{in}.

2.60 Repeat Problem 2.59 using Module 2.6.

*2.61 Using a slotted line on a 50 Ω air-spaced lossless line, the following measurements were obtained: $S = 1.6$ and $|\tilde{V}|_{max}$ occurred only at 10 cm and 24 cm from the load. Use the Smith chart to find Z_L.

2.62 At an operating frequency of 5 GHz, a 50 Ω lossless coaxial line with insulating material having a relative permittivity $\varepsilon_r = 2.25$ is terminated in an antenna with an impedance $Z_L = 150\ \Omega$. Use the Smith chart to find Z_{in}. The line length is 30 cm.

Section 2-11: Impedance Matching

*2.63 A 50 Ω lossless line 0.6λ long is terminated in a load with $Z_L = (50 + j25)\ \Omega$. At 0.3λ from the load, a resistor with resistance $R = 30\ \Omega$ is connected as shown in Fig. P2.63. Use the Smith chart to find Z_{in}.

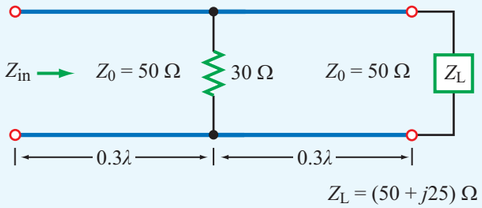


Figure P2.63 Circuit for Problem 2.63.

2.64 Use Module 2.7 to design a quarter-wavelength transformer to match a load with $Z_L = (100 - j200)\ \Omega$ to a 50 Ω line.

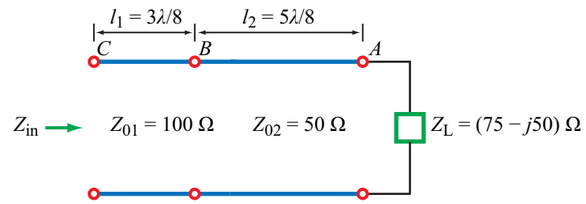


图 P2.50 问题 2.50 的电路。

*2.50 使用史密斯圆图确定图 P2.50 所示两线配置的输入阻抗 Z_{in}。

2.51 使用模块 2.6 重复问题 2.50。

*2.52 在终止于负载 $Z_L = 100\ \Omega$ 的无损传输线上，测得驻波比为 2.5。使用史密斯圆图找出 Z_0 的两个可能值。

2.53 无损 50 Ω 传输线的端接负载为 $Z_L = (50 + j25)\ \Omega$。使用史密斯圆图查找以下内容：

(a) 反射系数 Γ。

*(b) 驻波比。

(c) 负载 0.35λ 处的输入阻抗。 (d) 负载 0.35λ 处的输入导纳。 (e) 输入阻抗为纯电阻时的最短线路长度。 (f) 负载的第一个电压最大值的位置。

2.54 使用模块 2.6 重复问题 2.53。

*2.55 无损 50 Ω 传输线因短路而终止。使用史密斯圆图确定：

(a) 距负载 2.3λ 处的输入阻抗。 (b) 输入导纳为 $Y_{in} = -j0.04$ S 时距负载的距离。

2.56 使用模块 2.6 重复问题 2.55。

*2.57 如果 $z_L = 1.5 - j0.7$，则使用史密斯圆图求 y_L。

2.58 长度为 $3\lambda/8$ 的无损 100 Ω 传输线以未知阻抗终止。如果输入阻抗为 $Z_{in} = -j2.5\ \Omega$，

(a) 使用史密斯圆图求 Z_L。 (b) 使用模块 2.6 验证您的结果。

2.59 75 Ω 无损线的长度为 0.6λ。如果 $S = 1.8$ 且 $\theta_r = -60°$，则使用史密斯圆图求出 $|\Gamma|$、Z_L 和 Z_{in}。

2.60 使用模块 2.6 重复问题 2.59。

*2.61 使用 50 Ω 空气间隔无损线路上的开缝线，获得以下测量结果：$S = 1.6$ 和 $|\tilde{V}|_{max}$ 仅发生在距离负载 10 厘米和 24 厘米处。使用史密斯圆图求出 Z_L。

2.62 在 5 GHz 的工作频率下，采用相对介电常数 $\varepsilon_r = 2.25$ 的绝缘材料的 50 Ω 无损同轴线在阻抗 $Z_L = 150\ \Omega$ 的天线中端接。使用史密斯圆图求出 Z_{in}。线长为30厘米。

第 2-11 节：阻抗匹配

*2.63 一条 0.6λ 长的 50 Ω 无损线路终止于 $Z_L = (50 + j25)\ \Omega$ 的负载。在距负载 0.3λ 处，连接一个电阻为 $R = 30\ \Omega$ 的电阻，如图 P2.63 所示。使用史密斯圆图求出 Z_{in}。

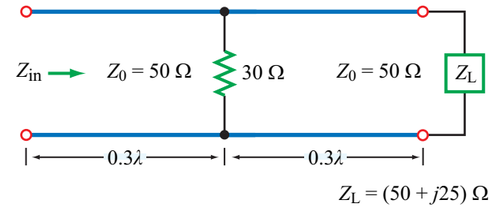


图 P2.63 问题 2.63 的电路。

2.64 使用模块 2.7 设计四分之一波长变压器，以将 $Z_L = (100 - j200)\ \Omega$ 的负载与 50 Ω 线路相匹配。

2.65 Use Module 2.7 to design a quarter-wavelength transformer to match a load with $Z_L = (50 + j10)\ \Omega$ to a 100 Ω line.

2.66 A 200 Ω transmission line is to be matched to a computer terminal with $Z_L = (50 - j25)\ \Omega$ by inserting an appropriate reactance in parallel with the line. If $f = 800$ MHz and $\varepsilon_r = 4$, determine the location nearest to the load at which inserting:

(a) A capacitor can achieve the required matching, and the value of the capacitor.

(b) An inductor can achieve the required matching, and the value of the inductor.

2.67 Repeat Problem 2.66 using Module 2.8.

2.68 A 50 Ω lossless line is to be matched to an antenna with $Z_L = (75 - j20)\ \Omega$ using a shorted stub. Use the Smith chart to determine the stub length and distance between the antenna and stub.

*__2.69__ Repeat Problem 2.68 for a load with

$$Z_L = (100 + j50)\ \Omega.$$

2.70 Repeat Problem 2.68 using Module 2.9.

2.71 Repeat Problem 2.69 using Module 2.9.

2.72 Determine Z_{in} of the feed line shown in **Fig. P2.72**. All lines are lossless with $Z_0 = 50\ \Omega$.

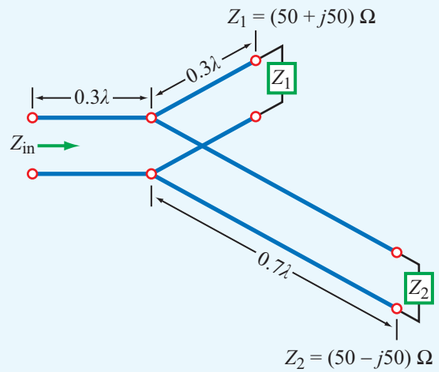


Figure P2.72 Network for Problem 2.72.

*__2.73__ Repeat Problem 2.72 for the case where all three transmission lines are $\lambda/4$ in length.

2.74 A 25 Ω antenna is connected to a 75 Ω lossless transmission line. Reflections back toward the generator can be eliminated by placing a shunt impedance Z at a distance l from the load (**Fig. P2.74**). Determine the values of Z and l.

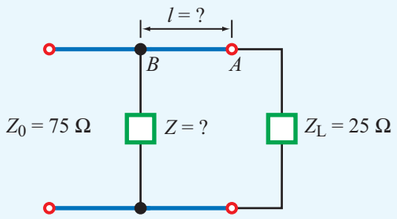


Figure P2.74 Circuit for Problem 2.74.

Section 2-12: Transients on Transmission Lines

2.75 Generate a bounce diagram for the voltage $V(z,t)$ for a 1-m long lossless line characterized by $Z_0 = 50\ \Omega$ and $u_p = 2c/3$ (where c is the velocity of light) if the line is fed by a step voltage applied at $t = 0$ by a generator circuit with $V_g = 60$ V and $R_g = 100\ \Omega$. The line is terminated in a load $R_L = 25\ \Omega$. Use the bounce diagram to plot $V(t)$ at a point midway along the length of the line from $t = 0$ to $t = 25$ ns.

2.76 Repeat Problem 2.75 for the current $I(z,t)$ on the line.

2.77 In response to a step voltage, the voltage waveform shown in **Fig. P2.77** was observed at the sending end of a lossless transmission line with $R_g = 50\ \Omega$, $Z_0 = 50\ \Omega$, and $\varepsilon_r = 2.25$. Determine:

(a) the generator voltage,

(b) the length of the line, and

(c) the load impedance.

*__2.78__ In response to a step voltage, the voltage waveform shown in **Fig. P2.78** was observed at the sending end of a shorted line with $Z_0 = 50\ \Omega$ and $\varepsilon_r = 4$. Determine V_g, R_g, and the line length.

2.65 使用模块 2.7 设计四分之一波长变压器，以将 $Z_L = (50 + j10)\ \Omega$ 的负载与 100 Ω 线路相匹配。

2.66 一条 200 Ω 传输线应通过与该线并联插入适当的电抗来与 $Z_L = (50 - j25)\ \Omega$ 的计算机终端相匹配。如果 $f = 800$ MHz 且 $\varepsilon_r = 4$，则确定最接近负载的位置：

(a) 一个电容器能否达到所需的匹配，以及电容器的值。
(b) 电感能否达到所需的匹配，以及电感的值。

2.67 使用模块 2.8 重复问题 2.66。

2.68 50 Ω 无损线路应使用短路短截线与 $Z_L = (75 - j20)\ \Omega$ 的天线相匹配。使用史密斯圆图确定短截线长度以及天线与短截线之间的距离。

*2.69 对负载重复问题 2.68

$$Z_L = (100 + j50)\ \Omega.$$

2.70 使用模块 2.9 重复问题 2.68。

2.71 使用模块 2.9 重复问题 2.69。

2.72 确定图P2.72所示进料线的Z_{in}。所有线都是无损的，$Z_0 = 50\ \Omega$。

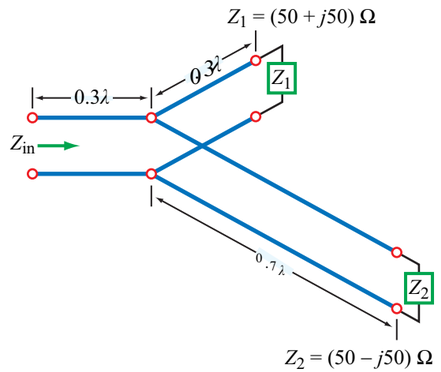


图 P2.72 问题 2.72 的网络。

*2.73 对于所有三个传输线长度均为 $\lambda/4$ 的情况，重复问题 2.72。

2.74 25 Ω 天线连接到 75 Ω 无损传输线。通过在距负载 l 处放置一个并联阻抗 Z 可以消除向发生器的反射（图 P2.74）。确定 Z 和 l 的值。

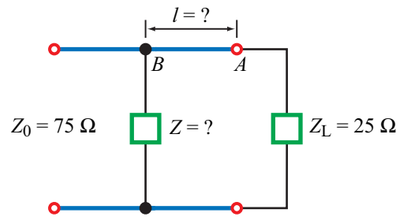


图 P2.74 问题 2.74 的电路。

第 2-12 节：传输线上的瞬变

2.75 如果由 $V_g = 60$ V 和 $R_g = 100$ 的发生器电路在 $t = 0$ 处施加阶跃电压，为 1 米长的无损线路生成电压 V (z,t) 的反射图，其特征为 $Z_0 = 50\ \Omega$ 和 $u_p = 2c/3$ （其中 c 是光速） Ω。该线路终止于负载 $R_L = 25\ \Omega$。使用反射图在从 $t = 0$ 到 $t = 25$ ns 的线长度中间的点处绘制 V (t)。

2.76 对线路上的电流 I(z,t) 重复问题 2.75。

2.77 响应阶跃电压，在无损传输线的发送端观察到如图 P 2.77 所示的电压波形，其中 $R_g = 50\ \Omega$、$Z_0 = 50\ \Omega$ 和 $\varepsilon_r = 2.25$。决定：

(a) 发电机电压，(b) 线路长度，(c) 负载阻抗。

*2.78 响应阶跃电压，在 $Z_0 = 50\ \Omega$ 和 ε_r = 短路线路的发送端观察到如图 P2.78 所示的电压波形。 4. 确定 V_g、R_g 和线路长度。

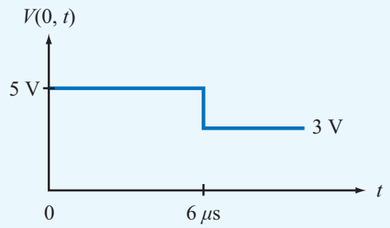


Figure P2.77 Voltage waveform for Problems 2.77 and 2.79.

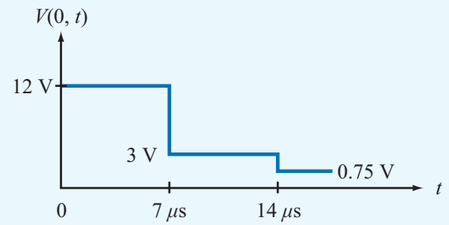


Figure P2.78 Voltage waveform of Problem 2.78.

2.79 Suppose the voltage waveform shown in **Fig. P2.77** was observed at the sending end of a 50 Ω transmission line in response to a step voltage introduced by a generator with $V_g = 15$ V and an unknown series resistance R_g. The line is 1 km in length, its velocity of propagation is 1×10^8 m/s, and it is terminated in a load $R_L = 100\ \Omega$.

- **(a)** Determine R_g.
- **(b)** Explain why the drop in level of $V(0,t)$ at $t = 6\ \mu$s cannot be due to reflection from the load.
- **(c)** Determine the shunt resistance R_f and location of the fault responsible for the observed waveform.

2.80 A generator circuit with $V_g = 200$ V and $R_g = 25\ \Omega$ was used to excite a 75 Ω lossless line with a rectangular pulse of duration $\tau = 0.4\ \mu$s. The line is 200 m long, its $u_p = 2 \times 10^8$ m/s, and it is terminated in a load $R_L = 125\ \Omega$.

- **(a)** Synthesize the voltage pulse exciting the line as the sum of two step functions, $V_{g_1}(t)$ and $V_{g_2}(t)$.
- **(b)** For each voltage step function, generate a bounce diagram for the voltage on the line.
- **(c)** Use the bounce diagrams to plot the total voltage at the sending end of the line.
- **(d)** Confirm the result of part (c) by applying Module 2.10.

2.81 For the circuit of Problem 2.80, generate a bounce diagram for the current and plot its time history at the middle of the line.

*__2.82__ In response to a step voltage, the voltage waveform shown in **Fig. P2.82** was observed at the midpoint of a lossless transmission line with $Z_0 = 50\ \Omega$ and $u_p = 2 \times 10^8$ m/s. Determine: (a) the length of the line, (b) Z_L, (c) R_g, and (d) V_g.

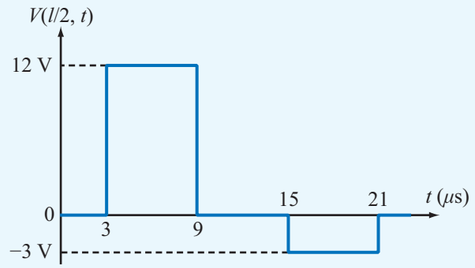


Figure P2.82 Circuit for Problem 2.82.

2.83 The transmission-line circuit of **Fig. P2.83** is excited by a 1-ns long pulse that starts at $t = 0$. The line is 1 m long and the phase velocity of the line is 2×10^8 m/s. Generate:

- **(a)** a plot of the voltage across the line at $t = 6$ ns,
- **(b)** a plot of the voltage across the line at $t = 12$ ns,
- **(c)** a plot of the voltage as a function of time at the sending end of the line, and
- **(d)** a plot of the voltage as a function of time at the midpoint of the line.

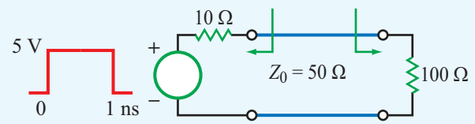


Figure P2.83 Circuit for Problem 2.83.

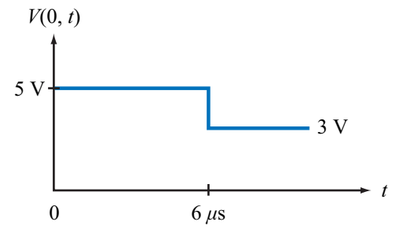


图 P2.77 问题 2.77 和 2.79 的电压波形。

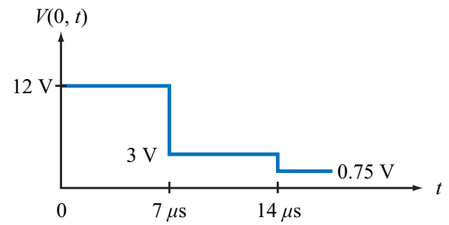


图 P2.78 问题 2.78 的电压波形。

2.79 假设在 50 Ω 传输线的发送端观察到图 P2.77 所示的电压波形，以响应 $V_g = 15$ V 和未知串联电阻 R_g 的发电机引入的阶跃电压。该线长 1 km，传播速度为 1×10^8 m/s，终止于负载 $R_L = 100\ \Omega$。

(a) 确定 R_g。 (b) 解释为什么在 $t = 6\ \mu$s 时 V (0,t) 电平的下降不可能是由于负载的反射造成的。 (c) 确定分流电阻 R_f 和导致观察到的波形的故障位置。

2.80 使用 $V_g = 200$ V 和 $R_g = 25\ \Omega$ 的发生器电路，用持续时间为 $\tau = 0.4\ \mu$s 的矩形脉冲激励 75 Ω 无损线。该线路长 200 m，其 $u_p = 2 \times 10^8$ m/s，终止于负载 $R_L = 125\ \Omega$。

(a) 将激励线路的电压脉冲合成为两个阶跃函数 $V_{g_1}(t)$ 和 $V_{g_2}(t)$ 之和。 (b) 对于每个电压阶跃函数，生成线路上电压的反弹图。 (c) 使用反弹图绘制线路发送端的总电压。 (d) 通过应用模块 2.10 确认 (c) 部分的结果。

2.81 对于问题 2.80 的电路，生成电流的跳动图，并在线的中间绘制其时间历史。

*2.82 响应阶跃电压，在 $Z_0 = 50\ \Omega$ 和 $u_p = 2 \times 10^8$ m/s 的无损传输线中点观察到图 P2.82 所示的电压波形。确定：(a) 线的长度、(b) Z_L、(c) R_g 和 (d) V_g。

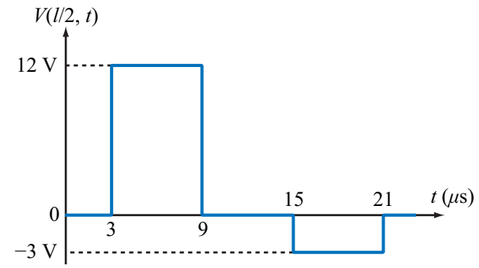


图 P2.82 问题 2.82 的电路。

2.83 图 P2.83 的传输线电路由从 t = 0 开始的 1 ns 长脉冲激励。传输线长 1 m，相速度为 2×10^8 m/s。产生：

(a) t = 6 ns 时线路两端电压的图，(b) t = 12 ns 时线路两端电压的图，(c) 线路发送端电压随时间变化的图，(d) 线路中点处电压随时间变化的图。

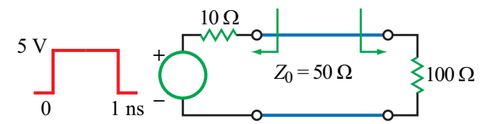
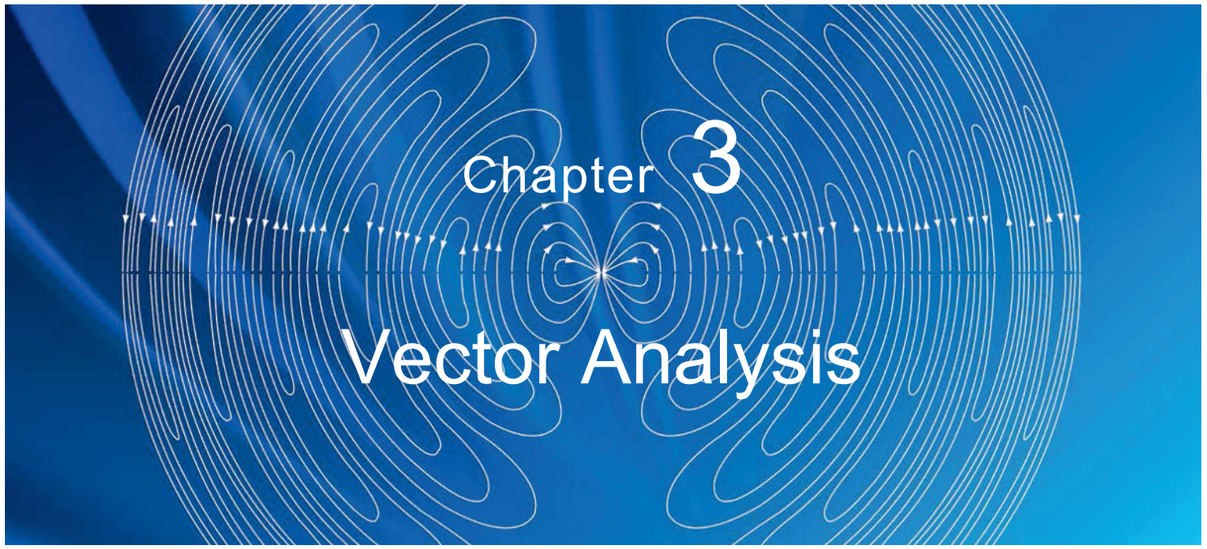


图 P2.83 问题 2.83 的电路。



Chapter Contents

Overview, 131
3-1 Basic Laws of Vector Algebra, 131
3-2 Orthogonal Coordinate Systems, 137
3-3 Transformations between Coordinate Systems, 143
3-4 Gradient of a Scalar Field, 147
TB5 Global Positioning System, 151
3-5 Divergence of a Vector Field, 153
3-6 Curl of a Vector Field, 157
TB6 X-Ray Computed Tomography, 159
3-7 Laplacian Operator, 162
Chapter 3 Summary, 163
Problems, 164

Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Use vector algebra in Cartesian, cylindrical, and spherical coordinate systems.
2. Transform vectors between the three primary coordinate systems.
3. Calculate the gradient of a scalar function and the divergence and curl of a vector function in any of the three primary coordinate systems.
4. Apply the divergence theorem and Stokes's theorem.

第3章

矢量分析

章节内容

概述，131 3-1 矢量代数基本定律，131 3-2 正交坐标系，137 3-3 坐标系之间的变换，143 3-4 标量场的梯度，147 TB5 全球定位系统，151 3-5 矢量场的散度，153 3-6 矢量场的旋度，157 TB6 X 射线计算机断层扫描，159 3-7 拉普拉斯算子，162 第 3 章总结，163 问题，164

目标

学习本章介绍的材料后，您应该能够：

1. 在笛卡尔坐标系、圆柱坐标系和球坐标系中使用矢量代数。 2. 在三个主坐标系之间变换向量。 3. 计算三个主坐标系中任意一个标量函数的梯度以及向量函数的散度和旋度。 4. 应用散度定理和斯托克斯定理。